MILITARY SPECIFICATION

PLASTIC SHEETS, POLYETHER URETHANE, FLEXIBLE

This specification is approved for use within Chemical Research, Development and Engineering Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers transparent, flexible sheets made of polyether urethane resin.

1.2 Classification. The sheets shall be furnished in nominal thicknesses of 0.060 inches.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

MILITARY

MIL-C-10758 - Chemical Agent GB.
MIL-C-12051 - Chemical Agent, Mustard, Distilled (HD).
MIL-D-50030 - Decontaminating Agent, DS2.

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, U.S. Army Chemical Research, Development and Engineering Center, ATTN: SMCCR-SPD-TS, Aberdeen Proving Ground, MD 21010-5423 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 9330

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STANDARDS

FEDERAL

Fed. Std. No. 406 - Plastic, Methods of Testing.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-282 - Filter Units, Protective Clothing, Gas-Mask Components and Related Products: Performance-Test Methods.

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

U.S. ARMY ARMAMENT, MUNITIONS AND CHEMICAL COMMAND

CHEMICAL RESEARCH, DEVELOPMENT AND ENGINEERING CENTER

D65540-A50-0089 - Lensometer, Q44.

(Copies of specifications, standards, handbooks, drawings, publications, an other Government documents required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted shall be those listed in the issue of the DODISS specified in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS shall be the issue of the non-government documents which is current on the date of the solicitation.

ASTM STANDARDS

D 570 - Test for Water Absorption of Plastics.
D 637 - Test for Surface Irregularities of Flat Transparent Plastic Sheets.
D 638 - Test for Tensile Properties of Plastics.
D 673 - Test for Mar Resistance of Plastics.
D 746 - Test for Brittleness Temperature of Plastics and Elastomers by Impact.
D 747 - Test for Stiffness of Plastics by Means of a Cantilever Beam.

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D 1003 - Test for Haze and Luminous Transmittance of Transparent Plastics.
D 1925 - Yellowness Index of Plastics.
D 2240 - Test for Rubber Property-Durometer Hardness.
D 3951 - Commercial Packaging.

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

(Nongovernment standards and other publications are normally available from the organizations which prepare or which distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specifications, specification sheets or MS standards), the text of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. The sheets shall be made of polyether urethane and whatever plasticizers, stabilizers and lubricants that are required to enable the finished product to satisfy the requirements of this specification.

3.2 Thickness, length, and width. The thickness of the sheets shall be within ± 10 percent of the nominal value (see 1.2). The minimum length and width of the sheets shall be as specified in the acquisition document (see 6.2).

3.3 Optical defects. Excluding the 1/2 inch band around the perimeter of the sheet, it shall be free from imbedded particles, scratches, bubbles, or other flaws which will reduce the visibility through the sheet material at approximately 1-1/2 inches from the eye. In addition, an area of 1 square foot may not contain more than two bubbles, imbedded particles, or spots with a maximum diameter of 1.0 mm or larger or four bubbles, imbedded particles, or spots with maximum diameter of 0.5 to 1.0 mm. All flaws of 0.5 mm or larger shall be at least four inches apart.

3.4 Physical and chemical requirements. The sheets shall meet the requirements of table I when tested in accordance with the applicable methods.

3.5 Preproduction. Unless otherwise specified in the contract or order, a preproduction sample shall be produced in accordance with this specification for examination and test (see 4.3).

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3.6 Workmanship. The sheets shall be colorless, smooth, and free from blemishes, striae, waves, and damage such as cracks, splits, and other visible defects which will impair their appearance or cause distortion of vision. In addition, the sheets shall be free from contamination.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Government's responsibility. The Government will be responsible for the performance of the tests in 4.4.4.2 and 4.4.4.3. Samples shall be forwarded to the laboratory designated by the contracting officer (see 6.2).

4.1.3 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5, for which specific inspection has not been provided in this specification, have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).

(b) Quality Conformance Inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of 50 square feet of material shall be manufactured and tested prior to regular production.

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4.3.2 Inspection procedures.

4.3.2.1 For examination and nondestructive tests. The sample polyether urethane shall be examined and tested in accordance with table I.

4.3.2.2 For destructive tests. A 40 square foot sample shall be used to supply the specimens for the destructive tests in table I.

4.3.3 Acceptance/rejection criteria. The preproduction samples shall meet the examination and tests specified by 4.3.2 to be acceptable. The contractor shall obtain written approval from the contracting officer before proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of sheets of one thickness, length, and width produced from one batch (see 6.3) of resin, under essentially the same conditions and offered for acceptance at one time.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105.

4.4.2.2 For destructive tests. A representative sample, 40 square feet minimum, shall be taken from each lot to supply the specimens for the lot acceptance tests specified in table I.

4.4.2.3 For GB and HD resistance. A representative sample, 10 square feet minimum, shall be taken from each lot and submitted to the Government for GB and HD tests (see 4.1.2).

4.4.3 Inspection procedure.

4.4.3.1 For examination and nondestructive tests. Sample sheets from each lot shall be examined in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 For destructive tests. Sample shall be tested as specified in table I.

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TABLE I. Requirements and tests

Property	Units	Requirement				Test method		Number of determination
		Unaged		Aged (4.4.4.4)		FED-STD or ASTM	Method or Para	
		Min	Max	Min	Max			
Brittle temperature	°F		-10		-10	ASTM D 746		6
Crease resistance	degree	90					4.4.4.1	10
Discoloration	%	85					4.4.4.7	5
GB resistance	minutes	240					4.4.4.2	5
HD resistance	minutes	240					4.4.4.3	5
Haze	%		4		4	ASTM D 1003		6
Mar resistance	%*		15			ASTM D 673		5
Distortion**	diopter		****				4.4.4.8	5
Stiffness in flexure at 75°F**	psi		10,000		10,000	ASTM D 747		6
Tensile strength	psi	1500		1500		ASTM D 638		6
Ultimate elongation	%	100		100		ASTM D 638		6
Water absorption at 75°F	%		1.5			ASTM D 570		3
Blocking			Slight			FED-STD-406	Method 1131 @ 38°C & 1 PSI	5
Weathering**	***		14 avg 16 ind			ASTM D 1925	4.4.4.5	10
Light transmission	%	85				ASTM D 1003		5
Hardness @ 75° + 5°F	Shore A	85				ASTM D 2240	4.4.4.6	5
Optical uniformity	Displacement factor		29			ASTM D 637	4.4.4.9	1
Resistance to DS2			*****				4.4.4.10	5

*Percent haze using 400 grams of No. 80 silicone carbide. Samples shall be dipped in water and wiped dry with a lint free cloth to remove all the silicone carbide before reading haze.

**For preproduction only.

***Yellowness Index shall be in accordance with ASTM D 1925.

****The difference in prismatic effect or refractive power may not exceed 1/8 diopter in any meridian.

*****A solid opaque circle shall not form on the material.

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4.4.3.3 Classification of defects.

(a) Plastic sheets, polyether urethane, flexible.

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
101		Optical defects (3.3)
102		Thickness, length, and width incorrect (3.2)
Minor:	AQL 2.5 percent defective	
201		Damage (3.6)
202		Contamination (grease or oil)

4.4.4 Tests. (See table I). When the same value of a requirement is to be met by sheets in both aged and unaged conditions, one-half of the number of determinations shall be made in each condition. When more than one determination of a property in a specified condition is made, the results shall be averaged before comparison with the requirement, unless otherwise noted.

4.4.4.1 Crease resistance. The crease resistance of the sheet shall be determined as follows, using specimens measuring 1 by 2 inches:

(a) Condition the specimens for 48 hours in an atmosphere maintained at a temperature of 75° ± 2°F and a relative humidity of 50 ± 2 percent.

(b) Fold each specimen to form a 1 inch square, and subject it to a uniform load of 25 pounds for 5 minutes.

(c) Remove the load and allow a 3 minute interval for free recovery; then immediately measure the dihedral angle formed at the crease.

4.4.4.2 GB resistance. GB (MIL-C-10758) resistance shall be determined in accordance with method T208 of MIL-STD-282. Test shall be run for a maximum time of 360 minutes if maximum cumulative amount of agent in the effluent is not reached.

4.4.4.3 HD resistance. HD (MIL-C-12051) vapor resistance shall be determined in accordance with method T209 of MIL-STD-282. Test shall be run for a maximum time of 360 minutes if maximum cumulative amount of agent in the effluent is not reached.

4.4.4.4 Aging. The specimens shall be conditioned for 150 hours in circulating air maintained at 160° ± 2°F and a relative humidity of 50 ± 5 percent. As an alternate, aging can be conducted for 24 hours at 212° ± 2°F with all other conditions the same. Failures at 212°F can be retested at 160° ± 2°F.

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4.4.4.5 Weathering. Specimens, one square foot each, shall be subjected to weathering for 72 ± 2 hours using an Atlas Weather-Ometer (type x-w) that is operated with the light cam only.

4.4.4.6 Hardness. Hardness shall be checked by plying up four layers of material and recording the maximum hardness reading.

4.4.4.7 Discoloration. After determining light transmission, in accordance with ASTM D 1003, the lens shall be aged in an air circulating oven at 155° ± 3°F for 24 continuous hours. The lens shall be removed and allowed to stand at room temperature for four hours, then tested again for light transmission and examined for compliance with the requirement in table I.

4.4.4.8 Distortion. Any region which, in the course of visual inspection or examination, shows evidence of distortion due to irregularity of either surface, inhomogeneity of plastic materials, surface or internal defects, shall be examined in the following manner: Measure the apparant prismatic effect and refractive power (using the Q44 Lensometer, D6540-A50-0089) and that of the region adjacent to the affected region, and check for compliance with the requirement in table I.

4.4.4.9 Optical uniformity. Test shall be performed as required by ASTM D 637. The sheet material shall be retained in a fixed flat orientation, so that bending shall not occur. The outer 1/2 inch perimeter need not be examined.

4.4.4.10 DS2 resistance. Place 0.1 ml DS2 (MIL-D-50030) on surface of a 1 by 2 inch specimen for 10 minutes. Rinse off DS2 with distilled water and observe for opacity.

4.4.5 Acceptance/rejection criteria. If any specimen or average of determinations, as applicable, fails to meet the applicable requirements of this specification, the lot represented shall be rejected.

5. PACKAGING

5.1 Commercial packaging. The sheets shall be prepared for shipment in accordance with ASTM D 3951.

6. NOTES

6.1 Intended use. The material covered by this specification is intended for use in the manufacture of chemical-biological mask lenses and related items.

6.2 Ordering data. Acquisition documents should specify the following:

(a) The title, number, and date of this specification.
(b) The sheet length and width required (see 3.2).

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(c) The laboratory to which samples should be sent for GB and HD testing (see 4.4.2.3).

(d) Preproduction:

(1) Time allowed for contractor submission of samples for Government test and evaluation after award of contract.

(2) Name and address of test facility and shipping instructions when testing is performed by the Government.

(3) Time required for Government to notify the contractor whether or not to proceed with production.

6.3 Batch. A batch is defined as that quantity of material which has been subjected to some unit chemical or physical mixing process intended to make the final product substantially uniform.

6.4 Subject term (key word) listing.

Lens material for chemical-biological mask
Flexible polyether urethane plastic sheets

6.5 Supersession data. This specification superedes Chemical Systems Laboratory Purchase Description EA-P-1313, dated 6 April 1982.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project No. 9330-AB13

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-51514(EA)	PLASTIC SHEETS, POLYETHER URETHANE, FLEXIBLE

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*: _____

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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